

Stochastic Modeling Notes

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Contents

1	Survival Models	5
1.1	Survival Function	5
1.2	The force of mortality	6
1.3	Actuarial notation	6

Chapter 1

Survival Models

1.1 Survival Function

Let (x) denote a life aged x , where $x \geq 0$. The death of (x) can occur at any age greater than x , and we model the future lifetime of (x) by a continuous random variable which we denote by T_x . This means that $x + T_x$ represents the age-at-death random variable for (x) . Let F_x be the distribution function of T_x , so that:

$$F_x(t) = Pr[T_x \leq t]$$

this is the **lifetime distribution**. On the other hand, the survival function is given by:

$$S_x(t) = 1 - F_x(t) = Pr[T_x > t]$$

$$S_0(x+t) = S_0(x)S_x(t)$$

We can interpret the probability of survival from birth to age $x+t$ as the product of:

- the probability of survival to age x from birth, and
- the probability, having survived to age x , of further surviving to age $x+t$.

If we know survival probabilities from any age $x > 0$, then we also know survival probabilities from future age $x+t$:

$$S_x(t+u) = S_x(t)S_{x+t}(u)$$

Any survival function for a lifetime distribution must satisfy the following conditions to be valid:

- $S_x(0) = 1$; the probability that a life currently aged x survives 0 years is 1.
- $\lim_{t \rightarrow \infty} S_x(t) = 0$, all lives eventually die
- The survival function must not be a non-increasing function of t .

1.2 The force of mortality

$$\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x)$$

Survival function in terms of mortality function:

$$S_x(t) = e^{-\int_x^{x+t} \mu_r dr}$$

1.3 Actuarial notation

$${}_t p_x = \Pr[T_x > t] = S_x \quad (1.1)$$

$${}_t q_x = \Pr[T_x \leq t] = 1 - S_x(t) = F_x(t) \quad (1.2)$$